

MicroRNA Expression Profiling in Colorectal Carcinoma: Identification and Validation of Novel Diagnostic, Prognostic and Predictive Biomarkers

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Abstract—This study aimed to investigate aberrant miRNA expression in colon cancer using a bioinformatics pipeline. miRNA expression and clinical data were retrieved from the National Cancer Institute's Cancer Genome Atlas Program (TCGA) database and were processed using DESeq2. 107 upregulated and 189 downregulated miRNAs are identified to be significantly correlated with the presence of colon cancer. Five of the most significant overexpressed and under-expressed miRNAs were discussed as potential colon cancer diagnosis and treatment biomarkers. The clinical implications of these findings are substantial, as the identified miRNAs could serve as non-invasive biomarkers for early detection and prognosis of colon cancer. Furthermore, these miRNAs may provide insights into the molecular mechanisms underlying colon cancer progression, potentially unveiling new therapeutic targets. Future studies should focus on validating these miRNAs in larger, independent cohorts and exploring their functional roles in colon cancer biology. This could pave the way for developing miRNA-based diagnostic tools and personalized treatment strategies, ultimately improving patient outcomes.

Keywords— *miRNA expression, colon cancer, epigenetics, bioinformatics, biomarkers*

I. Introduction

Colon cancer is the third most common cancer and second most deadly cancer worldwide, with approximately 1.8 million new cases and 900 thousand deaths annually [1]. Together with the aging population, a sedentary lifestyle and poor dietary habits increase the risk of colon cancer. The epidemiology of colon cancer varies significantly between different demographics and is affected by factors including risk factor exposure, genetic susceptibility, and genetic mutations, which affect the prognosis of the disease. The average survival rate for colon cancer is 5 years, which is still poor considering the advancement in molecular targeted therapies [2]. In the USA, approximately 145,600 new cases of CRC are

diagnosed annually, with 71% being colon cancers and 29% rectal cancers. Annually, about 50,000 Americans die from CRC. Between 2008 and 2014, the incidence of CRC decreased by 2.2% per year in men but remained stable in women. The mortality rate from CRC decreased annually by 1.8% in men and 1.4% in women from 1999 to 2015 [3]. Symptoms of CRC include rectal bleeding, abdominal mass, abdominal pain, change in bowel habits, unexplained weight loss, appetite loss, deep vein thrombosis, and iron-deficiency anemia [4].

Micro RNAs (miRNAs) are a class of highly conserved small non-coding RNAs that regulate genes post-transcriptional expression by binding with the 3' untranslated region of target mRNAs. miRNAs are involved in all fundamental cellular biological processes, which include metabolism, proliferation, differentiation, and apoptosis. Alterations in miRNA expression can disrupt target regulation and impact cellular homeostasis, contributing to diseases like cancer [5]. miRNAs are important in biomedical research as they can be potential disease biomarkers. Some studies suggest that miRNAs can directly interact with proteins and influence epigenetic mechanisms. Due to their consistency and detectability in blood, miRNAs may be non-invasive blood biomarkers. Therefore, understanding miRNA levels and their effects on mRNA regulation is crucial for disease diagnosis and treatment. However, there are difficulties in detecting miRNA, including short sequence lengths, variability in per-cell copy number, and sequence similarity within miRNA families, making it difficult for them to be analytical targets [6].

miRNAs play a massive role in various aspects of cancer, including cell cycle regulation, proliferation, apoptosis, angiogenesis, epithelial-mesenchymal transition (EMT), and tumor invasion. Dysregulation of miRNAs can alter the expression of genes associated with cancer progression, impacting both oncogenes and tumor

suppressors [7]. Huang et al. (2021) mentioned that miRNAs are involved in colon cancer development, progression, and metastasis. miRNAs have been found to interact with dysregulated signaling pathways in colon cancer, such as mitogen-activated protein kinases (MAPKs), Wnt, transforming growth factor- β (TGF- β), and EMT [2]. This study aims to investigate aberrant

II. MATERIAL & METHODS

A. Retrieval of miRNA Expression and Clinical Data

Gene expressions of colorectal cancer were retrieved from the National Cancer Institute's Cancer Genome Atlas Program (TCGA) database using the TCGAAbiolinks library in RStudio. A GCD query was built with the following parameters and arguments. The miRNA data were then obtained by executing the GCDdownload function on the GCDquery [9].

Parameter	Argument
Project	COAD
Data Category	miRNA-Seq
Data Type	miRNA Quantification Expression
Experimental Strategy	miRNA-Seq

B. Data Processing

1) Data Preparation

The downloaded data were prepared with the GCDprepare function and stored in a data frame. Row names are set to miRNA_ID for better identification, and column names are modified to remove prefixes and/or suffixes from sample IDs [10].

2) Metadata Alignment and Cleaning

A separate data frame consists of metadata fields, such as 'cases' and 'tissue_definition' to ensure accurate downstream analysis. These metadata fields were extracted from the dataset to represent each patient's specific condition and tissue type. The column names of the metadata data frame were set to match the column names of the miRNA expression data frame to ensure that the metadata of each sample corresponds to their expression profile. Samples with metastatic tissues are omitted from the data as they are considered outliers, with sample size insignificant to the whole dataset [11].

C. Differential Expression Analysis with DESeq2

A DESeqDataSet is then constructed using the previously matched miR_df and miR_meta by specifying the experimental design of '~ Condition' as the normal reference level. DESeq analysis is then conducted to determine the abnormally expressed miRNAs across all samples that contribute to the development of colon cancer. Log2fold change and p-value were used to discover the most differentially expressed miRNA [12].

D. Result Extraction and Visualization

Upregulated and downregulated miRNA expressions were extracted from the DESeq2 result by performing the get_upregulated and get_downregulated functions. The list of these upregulated and downregulated miRNA expressions was sorted from the highest log2fold change to the lowest before each being exported to a .tsv file. The counts for each miRNA expression are normalized and exported as a separate counts file. Based on the generated .tsv files, the volcano plot was chosen to visualize the data as it correctly and accurately shows the data points that were determined significant, especially regarding miRNA expression data. In the volcano plot itself, the x-axis represents the over

miRNA expression that correlates with the presence or absence of colon cancer in patient samples through the bioinformatics pipeline to pinpoint the list of miRNAs that significantly affect the development of colon cancer in the hope that they can be used as potential biomarker and therapeutic target [8].

or under-expression of the miRNAs, while the y-axis represents the fold expression of the miRNAs[8].

III. RESULTS AND DISCUSSION

Based on the previously mentioned pipeline, miRNA expression log2 fold change values were acquired. 107 upregulated and 189 downregulated miRNAs had log2 fold values significantly correlated with the presence of colon cancer. Log2 fold change means the log2 ratio of an expression's fold change compared to the whole dataset's mean.

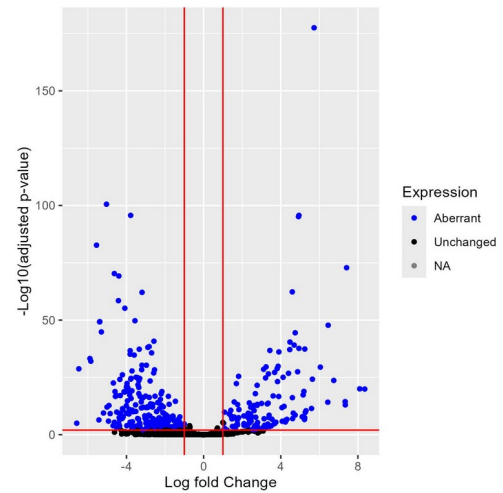


Figure 1. Volcano plot of log fold change and $-\log_{10}(\text{adjusted } p\text{-value})$. Absolute log fold change values above 1 are marked by the vertical red line, while the horizontal red line marks adjusted p-value above 0.01. With significant over or under-expressed miRNAs being highlighted as blue

As seen in Figure 1, the visualization of the different miRNA expressions showed how much miRNAs were over or under-expressed, which may cause the development of colorectal cancer.

Table 2. Five differentially expressed upregulated miRNA expressions with the highest log2 fold change. The absolute log2 fold change minimum value was set to 2

miRNA ID	Log2 Change	Fold	Standard Error of LFC	Wald Test Value	p-value
hsa-mir-135b	8.354456063		0.867804966	9.627112528	6.14E-22
hsa-mir-19a	8.093431509		0.838765066	9.649223413	4.95E-22
hsa-mir-374a	7.412549186		0.402009718	18.43873132	6.42E-76
hsa-mir-592	7.345833987		0.946322576	7.762505275	8.33E-15
hsa-mir-577	7.33936322		0.896561424	8.186124256	2.70E-16

Table 2 shows that the overexpression of miRNAs could be linked to the development of tumors and cancers. These excessive expressions have also been linked with studies showing that miR-135b interacts with the 3'UTR transcript of the APC gene, affecting mRNA levels and regulating APC expression, which participates in the tumorigenesis of colon cancer [13]. It has been proven that miR-19a suppressed ferroptosis in colon cancer by inhibiting the expression of the target gene IREB2 [14]. Overexpression of miR-374a has been shown to inhibit proliferation, invasion, migration, and intrahepatic metastasis of colon cancer cells in vitro and in vivo, while also inactive PI3K/AKT signaling, and downstream cell cycle, EMT factors [15]. miR-592 is suggested to act as an oncogene in colon cancer by directly inhibiting SPARC expression [16]. While miR-577 has been suggested to act as a tumor suppressor in colon cancer by targeting HSP27 [17], [18], [19].

Table 3. Five differentially expressed downregulated miRNA expressions with the highest log2 fold change. The absolute log2 fold

miRNA ID	Log2 Fold Change	Standard Error of LFC	Wald Test Value	p-value
hsa-mir-3622a	-6.579318997	1.372313	-4.79433	1.63E-06
hsa-mir-504	-6.470687259	0.558743	-11.5808	5.16E-31
hsa-mir-486-2	-5.894667301	0.473413	-12.4514	1.37E-35
hsa-mir-486-1	-5.849169456	0.478102	-12.2341	2.04E-34
hsa-mir-328	-5.554119666	0.282909	-19.6322	8.21E-86

change minimum value was set to 2

Table 3 highlights some miRNAs that were under-expressed significantly. These miRNAs were correlated with tumor suppressor genes, which elaborates their under-expression in the development of colon cancer. Previously done studies found that the has-mir-3622 is responsible for the management of silencing and suppressing imperfect base pairing of mRNAs by destabilizing the mRNA, as abnormal mRNA have also been responsible for tumorigenesis [20], [21], [22]. Meanwhile, hsa-mir-486-1 and hsa-mir-486-2, which are part of the hsa-miR-486 family, have been found to be responsible for degrading mRNA, which, in this case, as they are under-expressed, may explain the inability of these miRNAs to degrade imperfect mRNA which then leads to tumorigenesis [23]. The impact of hsa-mir-328 has been extensively explored before [24], [25] where it was found that miR-328 inhibited K562 cell proliferation, which is a chronic myeloid leukemia cell line, which means this significant underexpression possibly caused the tumorigenesis of colon cancer.

V. CONCLUSION

miRNA expression analysis through the bioinformatics pipeline is useful in identifying potential novel biomarkers for colon cancer. The identified miRNAs, such as has-mir-135 b, has-mir-19a, has-mir-374a, has-mir-592, and has-mir-577, were found to be significantly overexpressed in colon cancer samples. Conversely, miRNAs like hsa-mir-3622a, hsa-mir-504, hsa-mir-486-2, hsa-mir-486-1, and hsa-mir-328 were under expressed. These findings were in line with other studies on miRNAs' role in the development and progression of colon cancer. Further investigation into the functional implications of these miRNAs and how they interact with their target mRNA and affect the signaling pathway would help better understand their roles in colon cancer and verify their potential as biomarkers and therapeutic targets.

ACKNOWLEDGMENT

The authors thank the Department of Research and Community Service (LPPM) of the Indonesia International Institute for Life Sciences (i3L) for supporting this research. This paper has been improved with Grammarly Pro English editing software.

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